

DEPARTMENT OF AGRICULTURAL AND BIOENVIRONMENTAL ENGINEERING
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COURSE TITLE:

Drainage Technology (AES 423/AES 432)

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Drainage design Variables (LECTURE NO. 3)

In agricultural and civil drainage projects, effective planning begins with thorough site investigation. This process is carried out through a series of structured surveys, each of the surveys is meant to collect specific types of data that guide the design and implementation of an appropriate drainage system. These surveys are typically categorized into reconnaissance survey, preliminary survey, and design survey, carried out sequentially from the broadest overview to the most detailed engineering measurements.

Reconnaissance Survey

The reconnaissance survey is the first stage in drainage investigations and involves a broad, initial appraisal of the area to determine the general feasibility of drainage development. This type of survey is usually conducted through site visits, topographic maps, aerial photographs, satellite imagery, and interviews with local farmers or landowners. The aim is to identify major drainage problems, such as areas prone to waterlogging, flooding, or erosion, as well as the general slope, vegetation cover, and land use patterns. The reconnaissance survey helps to define the project boundaries and highlight any major obstacles or environmental concerns. It is qualitative in nature but may include rough elevation estimates, visual signs of soil saturation, and drainage outlet locations. For example, if a reconnaissance survey shows that a large portion of a field remains wet several days after rainfall, and there are no visible outflow points, it suggests a need for subsurface drainage and points to potential drainage constraints.

Preliminary Survey

Following the reconnaissance survey, the preliminary survey is conducted to collect quantitative data required for conceptual planning and comparison of alternative drainage layouts. This involves more detailed fieldwork, such as taking soil samples to determine texture, infiltration rates, and hydraulic conductivity; measuring water table levels. It also involves establishing elevation profiles using levelling instruments like dumpy levels or total stations. The preliminary survey also includes the measurement of slope gradients and natural drainage paths. These data are essential for assessing the suitability of different drainage methods (e.g., surface versus subsurface drainage) and for performing initial calculations of drainage coefficient, drain spacing, and system capacity. For instance, if the soil permeability were found to be less than 0.1 cm/hr during

the preliminary survey, it would indicate the need for closely spaced subsurface drains or even vertical drainage systems.

Design Survey

The design survey is the final and most detailed stage of investigation. Its purpose is to generate the precise engineering data needed to construct the drainage system. This survey includes high precision topographic mapping, detailed soil investigations at multiple depths, exact water table monitoring over time, and hydraulic testing of soil layers. The design survey often involves the setting of benchmarks, accurate longitudinal and cross-sectional profiling of ditches or drain lines, and determination of outlet capacity. Drainage modelling may also be applied using the collected field data. For example, using measured values of hydraulic conductivity and soil water retention, an engineer can use the Hooghoudt equation to determine the optimal drain spacing and depth. A typical design survey might conclude that subsurface drains should be placed at 1.2 meters depth and 30 meters spacing to effectively control the water table at a target depth of 1 meter during peak rainfall periods.

Soil Characteristics that soil drainage ^{is} affected

The flow of water in the soil (specifically as it pertains to soil drainage) is affected by the following factors –

1. Soil texture
2. Soil structure
3. Bulk density
4. Soil porosity
5. Hydraulic conductivity (K)
6. Soil matric potential
7. Depth to water table

Soil texture

Soil texture and soil structure are two fundamental physical properties that significantly influence the drainage behaviour of soil. Soil texture refers to the relative proportion of sand, silt, and clay particles in a soil. These particles vary in size: sand particles are the largest (0.05–2.0 mm), silt is intermediate (0.002–0.05 mm), and clay particles are the finest (<0.002 mm). The coarser the texture, the larger the pores and the faster the water can move through the soil profile. For instance, sandy soils have large macropores and typically exhibit high saturated hydraulic conductivity values, often between 5 to 20 cm/hr, which translates into rapid drainage. In contrast, clayey soils have smaller pores (micropores), leading to lower hydraulic conductivity—often less than 0.1 cm/hr—and poor drainage because water moves slowly.

Soil structure

Soil structure refers to the arrangement and aggregation of soil particles into peds (natural aggregates) or clumps. Good structure, such as granular or blocky structure, introduces macropores and inter-ped spaces, which significantly enhance water movement even in fine-textured soils. Poorly structured or compacted soils (e.g., massive or platy structures) restrict water movement due to a lack of continuous pores.

For example, assume a clay loam soil with good structure ($K_s = 0.5 \text{ cm/hr}$) and the same soil when compacted has a permeability coefficient of $K_s = 0.08 \text{ cm/hr}$. Assuming also that, both samples have the same cross sectional area for water flow of $A = 80 \text{ cm}^2$ and hydraulic gradient of $i = 0.3$, estimate the flow rate through each of the soil samples,

Recall that from Darcy's law that, $Q = K_s.A.i$

That is,

Q through the structured soil: $Q = K_s.A.i = 0.5 \text{ cm/hr} \times 80 \text{ cm}^2 \times 0.3 = 12 \text{ cm}^3/\text{hr}$

Q through the compacted soil: $Q = K_s.A.i = 0.08 \text{ cm/hr} \times 80 \text{ cm}^2 \times 0.3 = 1.92 \text{ cm}^3/\text{hr}$

Thus, from this example, it can be seen that **poor structure of the soil reduces drainage by over 80% even if the texture remains the same.**

This shows a five-fold reduction in water flow due to poor soil structure. Hence, even though texture defines the inherent pore size distribution, structure modifies the continuity and arrangement of those pores, greatly affecting drainage performance in the field.

In practice, these principles are critical in designing drainage systems. A soil with sandy texture may require minimal artificial drainage; while clayey or compacted soils may, need closely spaced tile drains or surface ditch systems. Furthermore, field practices such as avoiding soil compaction, adding organic matter, or deep ripping can enhance structure and thus improve drainage.

Bulk density

Bulk density is defined as the mass of dry soil per unit volume, including the air space or pores within the soil. It is a key indicator of soil compaction and is typically expressed in grams per cubic centimetre (g/cm^3) or kilograms per cubic meter (kg/m^3).

$$\rho_b = \frac{M_s}{V_t}$$

Where,

ρ_b = Bulk density (g/cm^3)

M_s = Dry soil mass (g)

V_t = Total soil volume (cm^3)

High bulk density reduces total pore volume, especially macropores; thereby reducing permeability.

A higher bulk density generally indicates a more compact soil with fewer pore spaces available for water movement. As bulk density increases, macropores—those large pores responsible for rapid drainage—are compressed or lost, reducing the soil's ability to transmit water.

Consider two loam soil samples,

Sample A: $M_s = 140g$, $V_t = 100\text{ cm}^3$; $\rho_b = 1.4g/cm^3$

Sample B: $M_s = 170g$, $V_t = 100\text{ cm}^3$; $\rho_b = 1.7g/cm^3$

Sample A will drain better due to lower compaction. In practical terms, K_s would be higher for sample A than for sample B since sample B more compacted as shown by its bulk density of $1.7g/cm^3$, which is lower than that of sample A.

In drainage design, soils with high bulk densities may require mechanical loosening (e.g., deep ripping or subsoiling) to restore adequate water movement.

Void ratio

Void ratio is another measure of soil pore space and is defined as the ratio of the volume of voids (air and water spaces) to the volume of solids:

$$V_e = \frac{V_v}{V_s}$$

Where,

V_e = Void ratio (dimensionless)

V_v = Volume of voids (cm^3)

V_s = Volume of solids (cm^3)

Although related to porosity, void ratio provides a clearer view of how pore space changes relative to the actual soil solid content, particularly in geotechnical applications. A higher void ratio indicates a looser soil structure with more space for water to flow. For example, a soil with a void ratio of 1.0 has equal volumes of solids and voids, facilitating good drainage, while a void ratio of 0.4 would suggest a more compact soil with limited drainage potential. As drainage depends on continuous pore pathways, a lower void ratio typically corresponds to more resistance to water flow, which can lead to surface ponding or waterlogging under rainfall or irrigation.

Porosity

Porosity is a direct determinant of how much water a soil can hold and transmit; it represents the volume fraction of soil that is pore space. It can be calculated using the following formula,

$$\phi = 1 - \frac{\rho_b}{\rho_s}$$

Where,

ϕ = Porosity

ρ_b = Bulk density (g/cm^3)

ρ_s = Solid density (g/cm^3)

However, not all pores contribute equally to drainage. Macropores (diameter > 0.08 mm) allow gravitational water to drain quickly, while micropores (diameter < 0.08 mm) retain water against gravity and slow down drainage. For example, two soils may have equal total porosity of 45%, but if one has more macropores due to good structure or biological activity, it will drain more rapidly. A well-structured loam with 15% macroporosity will drain far better than a clay with 5% macroporosity, even if both have the same total porosity. This difference significantly affects the soil's ability to remove excess water after rain or irrigation. Conversely, a fine-textured clay with high porosity but dominated by micropores will retain water longer and drain more slowly.

In summary, bulk density, porosity, and void ratio are interrelated physical properties that significantly affect soil drainage. Low bulk density, high total and macro-porosity, and high void ratio all favour better infiltration and water movement. Understanding and managing these properties is essential for effective drainage design, soil health, and crop productivity.

Hydraulic conductivity

Hydraulic conductivity is a measure of how easily water moves through soil pores under a hydraulic gradient. It can be classified into saturated and unsaturated hydraulic conductivity depending on the moisture condition of the soil. Saturated hydraulic conductivity, denoted as K_s refers to the rate at which water flows through a soil when all the pores are completely filled with water—that is, the soil is fully saturated. Under this condition, water movement is primarily driven by gravity, and the flow is typically rapid in sandy soils and very slow in clayey soils due to differences in pore size. The value of K_s is constant for a given undisturbed soil and can be determined using methods such as the constant head or falling head permeameter in the laboratory, or in situ techniques like the auger whole method in the field. In engineering applications, K_s is crucial for designing subsurface drainage systems, estimating infiltration rates, and analysing seepage through embankments and foundations.

For the estimation of the rate of flow of water in a given soil volume, Darcy's law can be used. The law states that, the rate of discharge per unit area of a given volume of soil is directly proportional to its hydraulic gradient. Mathematically, the Darcy's law is defined as,

$Q \propto A \cdot l \cdot i$

$Q = K \cdot A \cdot l$

Which is,

$Q = K_s \times A \times i$

Assuming that, $K_s = K$ and that $i = \frac{\Delta H}{L}$

Where,

Q = discharge

A = cross-sectional area

~~i = Hydraulic gradient.~~

K_s = Hydraulic conductivity of saturated soil (hydraulic conductivity/permeability coefficient of soil), cm/hr

K = Hydraulic conductivity/permeability coefficient of soil, cm/hr

i = Hydraulic gradient (dimensionless)

ΔH = Hydraulic head difference, m

L = Distance/ Length of flow, m

Example,

Suppose that we wish to compare the rate of flow of water through two soils, namely, sandy and clayey soils using identical conditions, that is, saturated conductivity values of $K_s = 10 \text{ cm/hr}$ (sandy soil), $K_s = 0.05 \text{ cm/hr}$ (Clayey soil).

Solution:

The rate of water flow through each of the soils calculated using the Darcy's law is given below,

For sandy soil: $Q = K.A.i = 10 \text{ cm/hr} \times 100 \text{ cm}^2 \times 0.05 = 500 \text{ cm}^3/\text{hr}$

For clayey soil: $Q = K.A.i = 0.05 \text{ cm/hr} \times 100 \text{ cm}^2 \times 0.05 = 2.5 \text{ cm}^3/\text{hr}$

This demonstrates that, under the same gradient and area, water flows about 99% times faster in sandy soil than in clayey soil, clearly showing how texture affects drainage.

Soil permeability can be determined either in the laboratory or in the field. In the laboratory, there are two methods for the determination of the said permeability coefficients and the choice of the method depends to a greater extent on the ease with which soil allows water to pass through it (i.e. soil type). For clayey soils that are characterised by lower permeability, the falling head permeameter is preferred. While for coarse-grained soils with high permeability such as sand, sandy-loam, and silty-loam, the constant head permeameter is preferred.

Falling head permeameter

The sketch of the falling head permeameter is shown below. It consists of a standpipe located at the top of the cylindrical soil sample container. Both of these are located on a water bath that has an outlet, which empties into a calibrated cylinder.

Assume that the standard pipe has an opening with a cross sectional area of a , assume also that as water flows from initial height of h_1 to its new height of h_2 , an elemental height of water contained in the pipe dh through an area of a forms an elemental volume of dV

To derive the equation for the estimation of the permeability coefficient for the falling head permeameter can be deduced as follows. First for the analysis of the flow of water through the soil, the Darcy's law is the starting point.

As stated earlier, $Q = K \cdot A \cdot i$ and from the statement on the elemental volume of water above, $dV = a \cdot dh$

Dimensionally, the discharge Q is in the units of cm/hr or in the dimensions of $Length/time = L/t$

Similarly, the rate of displacement of the elemental volume $\frac{dV}{dt} = Q$

Therefore, $Q = K \cdot A \cdot i$ becomes $\frac{dV}{dt} = K \cdot A \cdot i$

However, recall that, $i = \frac{h}{L}$

Thus, $\frac{dV}{dt} = K \cdot A \cdot \frac{h}{L}$

In the same manner, recall from the earlier statements on the elemental volume above that, $dV = a \cdot dh$

Therefore, $\frac{a \cdot dh}{dt} = K \cdot A \cdot \frac{h}{L}$

By rearrangement, the above equation can be written as,

$$\frac{dh}{h} = K \cdot A \cdot \frac{dt}{aL}$$

By integration and recalling that, $\int \frac{dh}{h} = h + c$ and that $\int dt = t + c$

Considering the limits for the definite integrals, it becomes

$$\int_{h_1}^{h_2} \frac{dh}{h} = \int_0^t K \cdot A \cdot \frac{dt}{aL}$$

$$\ln\left(\frac{h_1}{h_2}\right) = \frac{KA}{aL} t$$

Therefore, making K – the subject of the equation, it becomes

$$K = \frac{aL}{At} \ln\left(\frac{h_1}{h_2}\right)$$

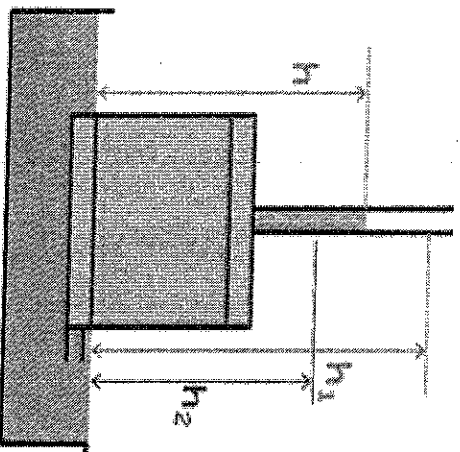


Figure 1. Falling head permeameter

Example 1:

A falling head test is conducted on a soil sample of the length 10 cm and cross-sectional area 30 cm². The standard has a cross-sectional area of 1 cm². The head drops from 40 cm to 10 cm in 300 seconds. Find the coefficient of permeability.

Solution:

Given that,

$$L = 10 \text{ cm}$$

$$A = 30 \text{ cm}^2$$

$$a = 1 \text{ cm}^2$$

$$h_1 = 40 \text{ cm}$$

$$h_2 = 10 \text{ cm}$$

$$t = 300 \text{ s}$$

Permeability coefficient,

$$K = \frac{1 \times 10}{30 \times 300} \ln \left(\frac{40}{10} \right) = \frac{10}{900} \ln(4) = 0.00111 \times 1.386 = 0.00154 \text{ cm/s}$$

Example 2:

A soil sample 20 cm long with a cross-sectional area of 50 cm² is tested using a standpipe of diameter 1.6 cm. The drops from 60 cm in 5 minutes. Compute the coefficient of permeability.

$$d = 1.6 \text{ cm}; r = 0.8 \text{ cm}$$

$$\text{Area of the soil sample, } a = \pi r^2 = 3.1416 \times 3.1416 \times 0.8^2 = 2.01 \text{ cm}^2$$

$$L = 20 \text{ cm}$$

$$A = 50 \text{ cm}^2$$

$$h_1 = 60 \text{ cm}$$

$$h_2 = 15 \text{ cm}$$

$$t = 5 \text{ minutes} = 300 \text{ sec.}$$

$$K = \frac{2.01 \times 20}{50 \times 300} \ln\left(\frac{60}{15}\right) = \frac{40.2}{15000} \ln(4) = 0.00268 \times 1.386 = 0.00371 \text{ cm/s}$$

Example 3:

A falling head test was conducted on a compacted clay sample 15 cm long and 25 cm² in cross-section. The water in a 0.5 cm² standpipe dropped from 100 cm to 20 cm in 10 minutes. Determine the coefficient of permeability.

Solution:

Given that,

$$L = 15 \text{ cm}$$

$$A = 25 \text{ cm}^2$$

$$a = 0.5 \text{ cm}^2$$

$$h_1 = 100 \text{ cm,}$$

$$h_2 = 20 \text{ cm,}$$

$$t = 10 \text{ minutes} = 600 \text{ sec.}$$

$$K = \frac{0.5 \times 15}{25 \times 600} \ln\left(\frac{100}{20}\right) = \frac{7.5}{15000} \cdot \ln(5) = 0.0005 \times 1.609 = 0.00080 \text{ cm/s}$$

Constant head permeameter

For the constant permeameter, the equation for the estimation of the coefficient of permeability can be deduced as follows,

According to the Darcy's law for the flow of water through a given of soil sample can be estimated as stated in the following equation,

$$Q = K \cdot A \cdot i$$

All variables in the equation above are as previously defined.

Recall also that the hydraulic gradient was previously defined as $i = \frac{h}{L}$

Where,

h = constant head difference

L = Length of the soil sample (cm)

Substituting the hydraulic gradient into the Darcy's equation above,

$$Q = K \cdot A \cdot \frac{h}{L}$$

By making the permeability coefficient the subject of the relationship in the above equation,

$$K = \frac{QL}{Ah}$$

Unlike the falling head permeameter, the constant head permeameter requires collecting a known volume of water and measuring the time taken for its discharge through the soil sample. Thus, assuming that V (cm^3) flows through the soil sample in time t (sec.) then, $Q = V/t$

Thus, substitute Q into the equation for K above,

$$K = \frac{QL}{Ah} = \frac{VL}{Aht}$$

Examples 1:

A soil sample 12 cm long with a cross-section area of 30 cm^2 is tested in a constant head permeameter. If 600 cm^3 of water flows through the sample in 5 minutes under a constant of 10 cm, compute the coefficient of permeability.

Solution:

Given the following data in this example,

$$L = 12 \text{ cm}$$

$$A = 30 \text{ cm}^2$$

$$V = 600 \text{ cm}^3$$

$$t = 5 \text{ minutes} = 5 \times 60 \text{ seconds/minute} = 300 \text{ sec.}$$

$$h = 10 \text{ cm}$$

$$\text{The permeability coefficient, } K = \frac{600 \times 12}{30 \times 10 \times 300} = \frac{7200}{90000} = 0.08 \text{ cm/s}$$

Example 2:

In a permeability test, water flows steadily through a sand sample 25 m long and 50 cm^2 . If 1000 cm^3 of water is collected in 2 minutes with a constant head of 20 cm, determine the coefficient of permeability.

Solution:

$$L = 25 \text{ cm}$$

$$A = 50 \text{ cm}^2$$

$$V = 1000 \text{ cm}^3$$

$$t = 2 \text{ minutes} = 2 \text{ minutes} \times 60 \text{ seconds/minute} = 120 \text{ sec.}$$

$$h = 20 \text{ cm}$$

$$\text{Permeability coefficient, } K = \frac{1000 \times 25}{50 \times 20 \times 10} = \frac{25000}{120000} = 0.2083 \text{ cm/s}$$

Example 3:

A clayey soil sample of length 10 cm and diameter 5 cm is tested under a constant head of 5 cm. In 10 minutes, 100 cm³ of water is collected. Calculated the coefficient of permeability.

Solution:

Given that,

$$L = 10 \text{ cm}$$

$$V = 100 \text{ cm}^3$$

$$t = 10 \text{ minutes} = 600 \text{ sec.}$$

$$h = 5 \text{ cm}$$

$$d = 5 \text{ cm; radius } r = 2.5 \text{ cm}$$

Therefore, the cross-sectional area of the sample, $A = \pi r^2 = 3.1416 \times 2.5^2 = 19.635 \text{ cm}^2$

$$\text{Permeability coefficient, } K = \frac{100 \times 10}{19.635 \times 5 \times 600} = 0.017 \text{ cm/s}$$

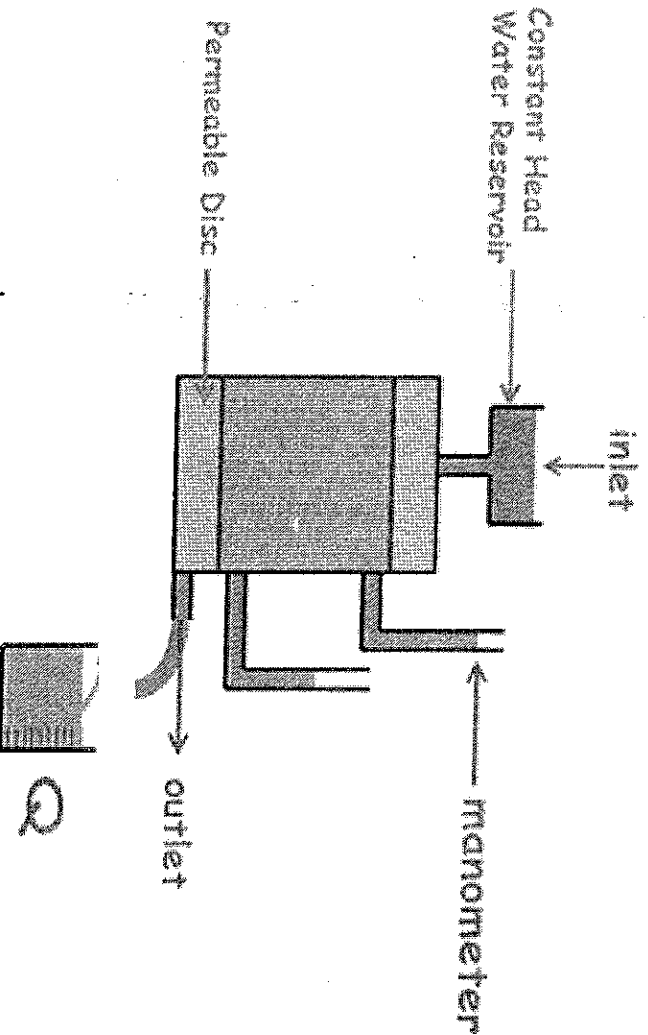


Figure 2. Constant head permeameter

Soil matric potential

Soil matric potential (also known as suction or capillary potential, (ψ_m) is a measure of the energy with which water is held by the soil matrix due to adhesion and capillary forces. Alternatively, it can be explained as the tension under which water is held in soil pores, usually negative in saturated soils. As the soil dries, the matric potential becomes more negative, and water moves slower; making it harder for water to drain under gravity. As such, soil matric potential plays a critical role in regulating the flow of water in unsaturated soils, significantly influencing soil drainage behaviour, especially above the water table.

In unsaturated conditions, water does not flow freely like in saturated soils. Instead, it moves from zones of higher (less negative) matric potential to lower (more negative) matric potential. This capillary-driven movement is relatively slow and depends on the continuity of water films between soil particles. For example, in a fine-textured clay soil with a matric potential of -100 kPa, water is held tightly in micropores and drains very slowly or not at all. In contrast, a sandy soil with a matric potential of -10 kPa drains more rapidly due to the dominance of macropores and weaker capillary forces.

Matric potential affects not only the rate but also the direction of water movement in soil. In sloped terrain or layered soils, water may move laterally if a sudden increase in soil moisture causes localized changes in matric potential. In drainage engineering, tensiometers and pressure plates are used to monitor matric potential to determine optimal times for irrigation or the need for subsurface drainage to remove excess water.

The unsaturated hydraulic conductivity, denoted as $K(\theta)$, is a related property that governs water flow under unsaturated conditions—when some pores contain air rather than water. As soil dries and volumetric water content (θ) decreases, the number and size of water-filled pores shrink, making water flow more difficult. Consequently, $K(\theta)$ is not constant but varies significantly with θ , and is always less than saturated hydraulic conductivity (K_s). Water movement under unsaturated conditions is influenced by both gravity and matric potential, and the relationship between water content and conductivity is nonlinear and complex. To address this complexity, empirical models such as the van Genuchten–Mualem model are used to relate $K(\theta)$ to the soil's moisture retention characteristics.

The Van Genuchten–Mualem model for unsaturated hydraulic conductivity is shown below,

$$K(\theta) = K_s S_e^\lambda \left[1 - \left(1 - S_e^{\frac{1}{m}} \right)^m \right]^2$$

Where,

$K(\theta)$ = Unsaturated hydraulic conductivity (cm/hr)

K_s = Saturated hydraulic conductivity (cm/hr)

S_e = Effective saturation

λ = Pore conductivity parameter (empirical constant, typically = 0.5)

$m = 1 - 1/n$; With $n > 1$

n = pore-size distribution index

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}$$

θ = Current volumetric water content (cm^3/cm^3)

θ_s = Saturation water content (cm^3/cm^3)

θ_r = Residual water content (cm^3/cm^3)

Example

Given a field with the following characteristics $K_s = 2.0 \text{ cm/hr}$, $\theta = 0.25 \text{ cm}^3/\text{cm}^3$, $\theta_s = 0.45 \text{ cm}^3/\text{cm}^3$, $\theta_r = 0.05 \text{ cm}^3/\text{cm}^3$, $n = 1.6$ and $\lambda = 0.5$, Compute $K(\theta)$

Solution:

The easiest approach for the computation of the $K(\theta)$ is to adopt sequential computation of the terms in the equation, as shown below,

Compute the following: m , S_e , S_e^λ , $S_e^{\frac{1}{m}}$, and $\left(1 - S_e^{\frac{1}{m}}\right)^m$

$$m = 1 - \frac{1}{n} = 1 - \frac{1}{1.6} = 0.375$$

$$S_e = \frac{0.25 - 0.05}{0.45 - 0.05} = \frac{0.20}{0.40} = 0.5$$

$$S_e^\lambda = 0.5^{0.5} = \sqrt{0.5} = 0.707$$

$$S_e^{\frac{1}{m}} = 0.5^{1/0.375} = 0.5^{2.667} = 0.151$$

$$\left(1 - S_e^{\frac{1}{m}}\right)^m = (1 - 0.151)^{0.375} = (0.849)^{0.375} = 0.949$$

Then plug all the computed terms into the Van Genuchten-Mualem model that is,

$$K(\theta) = K_s S_e^\lambda \left[1 - \left(1 - S_e^{\frac{1}{m}}\right)^m\right]^2$$

$$K(\theta) = 2.0 \times 0.707 \times [1 - 0.949]^2$$

$$K(\theta) = 2.0 \times 0.707 \times (0.051)^2$$

$$K(\theta) = 2.0 \times 0.707 \times 0.0026$$

$$K(\theta) = 0.0037 \text{ cm/hr}$$

From the above example, it can be seen that the unsaturated hydraulic conductivity (0.0037 cm/hr) is significantly lower than the saturated hydraulic conductivity of 2.0 cm/hr given in the question.

Measuring unsaturated hydraulic conductivity requires specialized equipment, such as tensiometers, pressure plate apparatus, or soil moisture sensors integrated into simulation models. Understanding $K(\theta)$ is

crucial for modelling water flow, planning irrigation schedules, and managing soil-water availability, particularly in rain fed or drip-irrigated systems.

For instance, if a soil has a matric potential of -33 kPa at field capacity and -1500 kPa at the wilting point, the unsaturated hydraulic conductivity between these points can drop by several orders of magnitude. Therefore, the soil drains rapidly when it is near saturation (less negative matric potential), but very slowly as it dries (more negative matric potential). This concept is fundamental to understanding drainage behaviour in the root zone and in designing controlled drainage systems that balance water retention with excess water removal.

Soil layering

Soil profiles with contrasting textures like sandy topsoil over a clayey subsoil can hinder vertical drainage due to the capillary barrier effect. For example, if the top soil is sandy layer ($K_s = 4$ cm/hr) and the sublayer is clay ($K_s = 0.1$ cm/hr), water will accumulate above the clay layer until the sandy layer is nearly saturated. This delays infiltration and creates temporary saturation. If rainfall adds 3 cm/hr, the upper layer may become saturated in a short time because water cannot pass into the sublayer faster than 0.1 cm/hr.

Depth of Water Table

A shallow water table reduces the ability of water to drain from the root zone. For instance, if the water table lies at 40 cm depth, the soil above it is saturated or nearly saturated, reducing the hydraulic gradient needed for downward movement. Capillary rise plays a role. For fine-textured soil with capillary pore radius $r = 0.00001$ m, capillary rise can be estimated by,

$$h = \frac{2\gamma \cos\theta}{\rho g r}$$

Where,

h = Capillary (m)

γ = Surface tension of water ($= 0.0728$ N/m at 20°C)

θ = Constant angle between water and soil ($= 0^\circ$ for most soils $\rightarrow \cos\theta \approx 1$)

ρ = Density of water (≈ 1000 kg/m³)

g = Acceleration due to gravity (≈ 9.81 kg/m³)

r = Capillary pore radius (m)